



OVERVIEW

Power BI provides a powerful tool called **Power Query Editor** for data cleaning and transformation. It is part of the ETL (Extract, Transform, Load) process that prepares your data for visualization and analysis.

DATA CLEANING PRINCIPLES IN POWER BI

Purpose:

Data cleaning ensures that the data is accurate, consistent, and usable. Dirty data leads to misleading visualizations and poor decision-making.

Common Data Cleaning Problems and How Power BI Helps

Problem	Example	Power BI Solution
Missing Values	Nulls or blanks in a column	Replace values, Fill up/down, Remove rows
Inconsistent Formatting	Dates in different formats, mixed text cases	Format using Transform tools
Duplicate Records	Same row appearing more than once	Remove duplicates
Trailing/Leading Spaces	" Abuja " vs "Abuja"	Trim and Clean functions
Typos and Misspellings	"Ngeria" vs "Nigeria"	Replace values manually or use fuzzy matching in merge
Incorrect Data	Numbers stored as text	Change data type to number, date,

Problem	Example	Power BI Solution
Types		etc.
Extra or Unnecessary Columns	System columns like RecordID , CreatedBy	Remove columns
Embedded Headers	Headers inside rows	Use “Use First Row as Headers” or promote headers
Blank Rows	Empty rows between data	Remove blank rows

🔄 DATA TRANSFORMATION PRINCIPLES IN POWER BI

🎯 Purpose:

Transformation reshapes data into the required structure for analysis. It includes splitting, merging, pivoting, aggregating, and more.

🔄 Common Data Transformation Needs and Solutions

Transformation Problem	Description	Power BI Feature/Solution
Unpivoted Data	Data in wide format	Use "Unpivot Columns"
Need to Aggregate	Summarize sales by region	Group By and Aggregate
Data in Multiple Tables	Need to join data from multiple sources	Merge Queries (Join), Append Queries (Union)

Transformation Problem	Description	Power BI Feature/Solution
One Column, Many Values	"Name: John, Age: 30" in one cell	Split Column by Delimiter
Complex Calculations	Need custom columns	Add Column > Custom Column
Irregular Date Structures	Financial year calendars, fiscal weeks	Add columns or transform date logic
Need for Lookup Tables	Replace codes with descriptions	Merge Queries or create relationships
Changing Column Headers Dynamically	Column names change with each refresh	Use dynamic M functions or custom logic
Creating Hierarchies	Region → Country → State	Create relationships or use grouping in Power BI model
Combining Files	Folder with multiple Excel/CSV files	Use "Folder" source and combine binaries

TEACHING STRATEGY

FOUNDATION (BASIC)

- Explain Power BI's Power Query interface.
- Import a messy Excel or CSV file.
- Demonstrate trimming, replacing, and formatting.

Classwork:

- Clean a file with missing values and typos.
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INTERMEDIATE

- Teach joins with “Merge Queries”.
- Demonstrate “Unpivot Columns” and “Group By”.
- Transform a date column and split text.

Classwork:

- Combine two datasets (e.g., Sales and Region).
 - Unpivot monthly sales columns into rows.
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ADVANCED

- Use custom columns and conditional columns.
- Automate folder loading (e.g., daily reports).
- Handle dynamic headers using M code.
- Resolve mismatched schemas between files.

Classwork:

- Load multiple Excel files, unify them, and clean the combined data.
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TOOLS TO HIGHLIGHT IN POWER QUERY

- **Transform Tab** (Data types, replace, format)
 - **Add Column Tab** (Custom, index, conditional)
 - **Home Tab** (Remove rows/columns, merge, append)
 - **View Tab** (Query Dependencies, Advanced Editor)
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CLASS NOTE: Data Cleaning and Transformation in Power BI

Title: Mastering Data Cleaning & Transformation Using Power BI

Tool: Power BI Desktop

File: `PowerBI_Cleaning_and_Transformation_Examples.xlsx`

CLASS OBJECTIVES

By the end of this class, students should be able to:

- Identify and fix common data quality issues.
 - Perform essential transformations for effective data modeling.
 - Use Power Query Editor in Power BI for cleaning and reshaping data.
 - Understand and use functions like remove duplicates, split, unpivot, merge, and more.
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CLASS STRUCTURE

- **Module 1:** Introduction to Power BI and Power Query
 - **Module 2:** Data Cleaning (Using `DataCleaning` sheet)
 - **Module 3:** Data Transformation (Using `DataTransformation` sheet)
 - **Module 4:** Recap + Class Project + Q&A
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MODULE 1: Introduction to Power BI and Power Query

Step-by-Step:

1. Open Power BI Desktop
2. Click on "Get Data" → Choose Excel → Load
`PowerBI_Cleaning_and_Transformation_Examples.xlsx`
3. Select the sheet `DataCleaning` → Click "Transform Data"

This opens the **Power Query Editor** where cleaning and transformation are done.

MODULE 2: DATA CLEANING — Practical with `DataCleaning` Sheet

Issue 1: Trailing/Leading Spaces

- **Column:** `Name` and `Region`
 - **Steps:**
 - Select the `Name` column → Home tab → *Transform* → Format → **Trim** and **Clean**
 - Do same for `Region`
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Issue 2: Inconsistent Capitalization

- **Column:** `Region`
 - **Steps:**
 - Select `Region` → Transform tab → Format → **Capitalize Each Word** or **Lowercase**
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🔍 Issue 3: Duplicate Records

- **Column(s):** Entire Rows
 - **Steps:**
 - Home tab → Remove Rows → **Remove Duplicates**
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🔍 Issue 4: Missing Values

- **Column:** Email , ID , Name
 - **Steps:**
 - Home tab → Keep Rows → **Remove Blank Rows**
 - Optional: Replace missing values → Transform → Replace Values → Replace null with "Unknown"
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🔍 Issue 5: Typos & Wrong Data

- **Column:** Name (e.g., "Ngeria" → "Nigeria")
 - **Steps:**
 - Right-click cell → Replace Values → "Ngeria" → "Nigeria"
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🔍 Issue 6: Wrong Data Types

- **Column:** Age , Salary
 - **Steps:**
 - Change column type from "Any" to "Whole Number" or "Decimal Number"
 - Use "Detect Data Type" for quick conversion
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🔍 Issue 7: Inconsistent Date Formats

- **Column:** JoinDate

- **Steps:**
 - Change data type to Date
 - Use "Using Locale" if needed
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❓ **MODULE 3: DATA TRANSFORMATION — Practical with DataTransformation Sheet**

↻ **Task 1: Split Full Name into First and Last**

- **Column:** FullName
 - **Steps:**
 - Select FullName → Transform → Split Column → By Delimiter → Choose ,
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↻ **Task 2: Unpivot Monthly Sales Columns**

- **Columns:** Jan Sales , Feb Sales , Mar Sales
 - **Steps:**
 - Select these three columns → Transform tab → **Unpivot Columns**
 - Rename new columns: Month , Sales
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↻ **Task 3: Standardize Region Codes**

- **Column:** Region Code
 - **Steps:**
 - Prepare a lookup table (e.g., NG → Nigeria, GH → Ghana)
 - Use "Merge Queries" to join Region Code with lookup
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🔄 Task 4: Create Department Relationships

- Use `DepartmentID` column to join with external Department table in your model later.
 - For now, rename and format correctly.
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🔄 Task 5: Change Data Types

- Ensure `EmployeeID` , `Sales` , `DepartmentID` are correct data types.
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🎓 MODULE 4: CLASS PROJECT + RECAP

💡 Project Idea:

"Clean and Transform an HR Dataset"

- Students will perform trimming, replacing, splitting, merging, and unpivoting on a real dataset using Power Query.
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✓ CLASSWORK AND ASSESSMENT

Task	Description
🔍 Clean <code>DataCleaning</code> sheet	Trim spaces, remove duplicates, fix typos, and replace missing values
🔄 Transform <code>DataTransformation</code> sheet	Split names, unpivot sales, join lookup table
🔍 Bonus	Create a custom column that labels high vs low sales
